DatumIQ

Product Requirements Document (PRD)
The Autonomous AI Chief Data Officer

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1. Executive Summary

DatumIQ is an enterprise-grade, Multi-Agent AI system designed to function as an autonomous **Chief Data Officer (CDO)**. Modern enterprises are drowning in unstructured and structured data silos, yet they lack the specialized engineering resources to extract timely, optimal business decisions. DatumIQ bridges this gap by enabling non-technical stakeholders to orchestrate data validation, processing, multi-modal analysis, visual synthesis, and strategic executive reporting using natural language. Built upon a specialized mesh of LLM-backed autonomous agents, DatumIQ delivers comprehensive, trace-auditable business recommendations alongside reproducible analytical code codeblocks within minutes.

2. Product Vision & Value Proposition

2.1 Product Vision

To democratize executive-level data strategy, shifting the corporate paradigm from human-dependent, weeks-long data analysis cycles to real-time, objective, and multi-agent derived synthesis.

2.2 Business Value Proposition

- **Cost Reduction:** Shrinks operational reliance on extensive data engineering and data analytics teams for standard ad-hoc analytical queries.
- **Velocity:** Reduces time-to-insight from standard 5–10 business days to under 180 seconds.
- **Accuracy & Security:** Eliminates human hallucination and structural calculation errors through automatic containerized code execution and strict semantic schema confirmation.

3. Problem Statement & Market Opportunity

3.1 Problem Statement

Mid-to-large market businesses face an acute data bottleneck. Raw data ingestion is fragmented across CSV formats, local Excel workbooks, and cloud data warehouses. Current Self-Service BI utilities (e.g., Tableau, PowerBI) require complex upfront semantic modeling and deep mastery of querying languages (SQL, DAX). Consequently, operational executives make multi-million dollar bets based on incomplete intuition or stale weekly spreadsheets.

3.2 Market Opportunity

The convergence of Enterprise AI spend and the democratization of Large Language Models has opened a massive greenfield market in the data analytics stack. By targeting non-technical enterprise decision-makers with a turnkey, zero-configuration agent platform, DatumIQ intercepts traditional BI consulting verticals, addressing a multi-billion dollar market segment spanning operational logistics, SaaS churn optimization, and financial forecasting.

4. Target Users & Personas

User Persona	Core Pain Points	DatumIQ Application Value
Sarah (VP of Sales & Operations)	Cannot wait 2 weeks for data teams to build Custom Cohort reports; data changes rapidly.	Direct natural language querying, instant executive PDF summaries for leadership alignment.
David (Director of Finance)	Data files often arrive with missing fields, syntax errors, or mismatched formatting across quarters.	Automated structural cleansing and cross-validation handled by the Data Validation Agent.
Alex (VP of Data Engineering)	Exhausted by simple, ad-hoc query requests; highly protective over PII leakages and raw database access.	Strict Sandboxed execution, local enterprise data privacy filters, and fully auditable execution paths.

5. Multi-Agent System Architecture

DatumIQ moves away from brittle, monolithic chain-of-thought processing. It relies on a deterministic, collaborative multi-agent mesh coordinated by a central planning protocol.

1. Planner Agent

Deconstructs user statements into a multi-step execution Directed Acyclic Graph (DAG). Monitors agent task fulfillment, performs rollback operations if execution branches error out, and updates the task blackboard state dynamically.

2. Data Validation Agent

Performs automated semantic checks. Profiles columns, maps missing anomalies, verifies data types, evaluates statistical variance, and handles missing row interpolations safely before execution begins.

3. Security & Privacy Agent

Enforces dynamic masking of PII elements (SSN, credit card data, hashes). Audits dynamically generated code blocks for external network calls and potential malicious command injections.

4. Analysis Agent

Translates logical sub-queries into algorithmic Python execution strings. Operates in a secure, stateless gRPC-connected sandbox, leveraging NumPy, Pandas, and SciPy to process clean datasets.

5. Visualization Agent

Evaluates categorical and continuous features to construct relevant visual representations (e.g., Matplotlib, Plotly). Automatically adjusts contrast, colors, labels, and layouts to professional presentation standards.

6. Recommendation Agent

Acts as the strategic domain layer. Translates empirical mathematical correlations into functional tactical initiatives (e.g., identifying optimization metrics, advising capital adjustments).

7. Report Agent

Assembles individual insights into structural formats. Generates professional HTML structures compiled natively via WeasyPrint into highly polished Executive PDFs or clean interactive markdown outputs.

6. Functional Requirements

6.1 Core Data Ingestion & Connectors

- The system MUST support file ingestion up to 500MB via raw CSV, Parquet, and Excel (.xlsx) file uploads.
- The system MUST provide native write-isolated database driver layers for Snowflake, Google BigQuery, PostgreSQL, and AWS Redshift.

6.2 Natural Language Processing & Query Mapping

- The system MUST decode semantic queries (e.g., "Calculate our 90-day LTV trends grouped by acquisition cohorts") without requiring explicit schema naming inputs.

6.3 Isolated Sandbox Execution Engine

- All algorithmic execution loops MUST occur within secure, isolated micro-containers.
- The internal environment MUST lack external network routing layers to prevent outbound malicious exfiltration of business states.

7. Non-Functional Requirements

- **Latency & Performance:** Sub-query validation models must execute within $T \leq 1.5s$. End-to-end orchestration pipelines (parsing to final PDF generation) must complete within $T \leq 180s$.
- **Scalability:** Supporting concurrent scheduling pipelines processing multi-gigabyte data partitions across a horizontally scalable Redis-backed task queue.
- **Maintainability:** Complete trace auditing available for each intermediate step of the generated analytical logic.

8. Security, Privacy, & Compliance

> **Enterprise Security Standard Compliance**
> All data stored in transit must utilize TLS 1.3 protocol architectures, with static structures hardened via AES-256 block ciphers. The system maintains strict adherence to SOC2 Type II, GDPR, and HIPAA data processing restrictions.

Data scrubbing operations run autonomously during the ingestion layer. PII objects are actively stripped or tokenized into local environment hashes. Because the analysis runtimes execute entirely inside a stateless container sandbox, data traces are instantly flushed out upon user session timeout tokens, preventing data pollution across multi-tenant infrastructures.

9. Technical Stack Strategy

- **LLM Inferences & Orchestration:** Gemini 1.5 Pro / Ultra APIs (via Vertex AI) for planning and report generation; customized fine-tuned CodeGemma parameters for high-throughput analytical query loops.
- **Backend Runtimes & Pipelines:** FastAPI, Python 3.11+, Celery task distribution systems backed by resilient Redis brokers.
- **PDF Generation Layer:** WeasyPrint headless engines compiling standards-compliant semantic HTML layouts with absolute programmatic layout formatting.

10. MVP Scope vs. Future Roadmap

Milestone Target	Included Features & Agent Capabilities	Target Timeline
Phase 1: MVP Release	Flat-file support (CSV/XLSX), core Planner, Validation, Analysis, and Report agents. Local containerized processing, structured PDF exports.	Day 0 - Day 90
Phase 2: Enterprise Integration	Live database connection strings (BigQuery, Snowflake). Integration of the Security Agent for real-time data masking. Slack/Teams alerting loops.	Day 90 - Day 180
Phase 3: Advanced Cognitive CDO	Predictive Time-Series modeling, autonomous multi-modal image ingestion, proactive optimization triggers based on real-time data shifts.	Day 180+

11. Competitive Analysis

- **Traditional BI (Tableau / PowerBI):** Heavy resource configurations, steep learning curves, entirely descriptive analytics lacking native recommendation workflows.
- **Standard LLM Chatbots (ChatGPT / Claude Advanced):** Limited context windows for massive data arrays, frequent code generation syntax errors, and data privacy concerns with public fine-tuning loops.

- **DatumIQ Advantage:** End-to-end automated orchestration, zero manual syntax building, secure micro-sandboxing, and strategic business recommendations tailored to executive requirements.

12. Success Metrics (KPI Framework)

The success of the platform deployment across market segments will be judged against three pillars:

1. **Task Completion Success Rate (TCSR):** $TCSR \geq 95\%$ of planning DAGs must run to completion without unhandled exceptions or code block restarts.
2. **Mean Time to Insight (MTTI):** Under 3 minutes for complex operational queries spanning multiple large datasets.
3. **User Retention & Engagement:** Month-over-Month growth in query generation volume per active enterprise client.

13. Risk Assessment & Mitigation Strategies

- **Risk: Code Generation Halting.** Complex data structures can trigger edge cases that break Python execution blocks. *Mitigation:* The Planner Agent monitors traceback loops, allowing the Analysis Agent up to two automated corrections before gracefully returning fallback insights.
- **Risk: Hallucinated Contextual Advice.** The Recommendation Agent could infer flawed market trajectories. *Mitigation:* Ground every insight strictly in mathematical variables derived from verified data blocks, completely banning non-supported conceptual inferences.